

PROPOSAL PROJECT
DIGITAL IMAGE PROCESSING
S1 DATA SCIENCE - UNIVERSITAS NEGERI SURABAYA

PROJECT IDENTITY	
Title	Robustness Analysis of Casting Defect Detection System Against Gaussian Noise Using HOG, LBP Feature Extraction and Support Vector Machine
Topic	Defect Identification
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1. INTRODUCTION
1.1. Background
The metal manufacturing industry, particularly the casting process, is a production sector that relies heavily on quality inspection to ensure that no defective products make it to the next stage. Defects in castings, such as blow holes, surface cracks, and sand inclusions, can lead to component failure in the field, resulting in financial losses and safety risks [1]. Traditionally, quality inspection has been performed manually by operators, but this approach has proven slow, inconsistent, and unscalable for modern production lines [2]. Digital image processing-based quality inspection automation offers a more reliable solution. The combination of conventional feature extraction techniques such as Histogram of Oriented Gradients (HOG) and Local Binary Pattern (LBP) has proven effective for detecting defects in manufactured metallic objects [3]. HOG excels at capturing structural and edge information that represents shape anomalies, while LBP complements it by capturing micro-surface texture patterns such as porosity [4]. These combined features can then be classified using a Support Vector Machine (SVM), known to be effective for high-dimensional classification problems with limited data [5]. However, a fundamental issue often overlooked in automated inspection research is the domain gap between training conditions and real-world deployment conditions. Models trained on clean data in controlled environments often experience significant performance degradation when deployed on the factory floor, where camera sensors exposed to high temperatures, machine vibrations, and dust gradually introduce noise into the images [6]. Gaussian noise, the most common form of degradation due to the limitations of sensor electronics, directly corrupts the gradient information on which HOG descriptors operate [4]. Most model evaluation studies only test performance on well-curated datasets, without considering the limitations of deployment conditions [1]. One proposed strategy to address this issue is noise-based data augmentation during training. By exposing the model to noisy data in a controlled manner, it is hoped that the model will gain greater resilience to similar perturbations during inference [7][11]. However, noise-based augmentation also risks degrading the model's ability to recognize clean images, so the trade-off needs to be systematically assessed [8].

Based on these problems, this study formulates the main questions: to what extent is the robustness of the HOG+LBP+SVM-based casting defect detection system to Gaussian noise, and can noise-based data augmentation improve the robustness of the system without sacrificing performance under ideal conditions?

1.2. Problem Solving Approach

To answer these research questions, this study proposes a robustness analysis methodology based on four systematic cross-testing scenarios using a submersible pump impeller casting product dataset from a real industrial environment [1].

The research pipeline begins with feature extraction using a combination of HOG and LBP from uniformly sized grayscale images. HOG extracts a histogram-based descriptor of gradient orientations at local cells capable of spatially capturing defect structures, while LBP generates a histogram of local binary patterns sensitive to surface texture variations [3]. The combined features are standardized using StandardScaler before being fed to an SVM classifier with an RBF kernel [5].

Image degradation is simulated by injecting Gaussian noise with a specified standard deviation into the entire dataset, resulting in two versions of the data: clean and noisy. Two SVM models are trained separately, resulting in four evaluation scenarios: (1) Clean Model → Clean Test as an ideal performance baseline; (2) Clean Model → Noise Test to assess the vulnerability of conventional systems; (3) Noise Model → Clean Test to verify that augmentation does not impair generalizability; and (4) Model Noise → Test Noise to prove the benefits of augmentation in strengthening model robustness [7][8]. The evaluation was conducted using accuracy, recall, and F1-score metrics accompanied by a confusion matrix for each scenario, resulting in a comprehensive trade-off analysis as a basis for recommendations for deploying inspection systems in real industrial environments [9].

2. Methodology

2.1. Dataset Exploration

The dataset used is the Real-Life Industrial Casting Product Dataset obtained from the Kaggle platform [1]. This dataset contains images of casting products, such as submersible pump impellers, collected directly from real industrial environments, so the available data reflects the actual conditions of the production process, not laboratory simulations.

This dataset is divided into two main subsets: training data and test data. Each subset contains two class categories: `def_front`, which represents defective products, and `ok_front`, which represents products in normal condition. Overall, this dataset contains approximately 6,633 images in the training data and 715 images in the test data. The proportion of classes in this dataset is fairly balanced between defective and normal classes, so no special treatment is required for class imbalance.

In terms of visual characteristics, images in the defect class generally display uneven product surfaces, characterized by the presence of anomalies such as blow holes (air cavities), surface cracks, and sand inclusions that appear as irregular patterns in certain areas [4]. Meanwhile, images in the normal class display relatively uniform and clean surfaces without any significant visual disturbances. All images in this dataset were captured from a frontal view of the product against a relatively uniform

background, allowing for differences between classes to be identified primarily through the texture and edge patterns on the product's surface. Each image will be loaded using OpenCV and resized to 128x128 pixels to ensure uniform dimensions for all data before proceeding to the next processing stage. Prior to entering the main pipeline, an Initial Data Exploration (EDA) will be performed, providing a baseline for understanding the data prior to the degradation and feature extraction processes.

2.2. Research Steps

This research will be conducted through five main stages, arranged systematically and sequentially, as illustrated in the following flowchart.

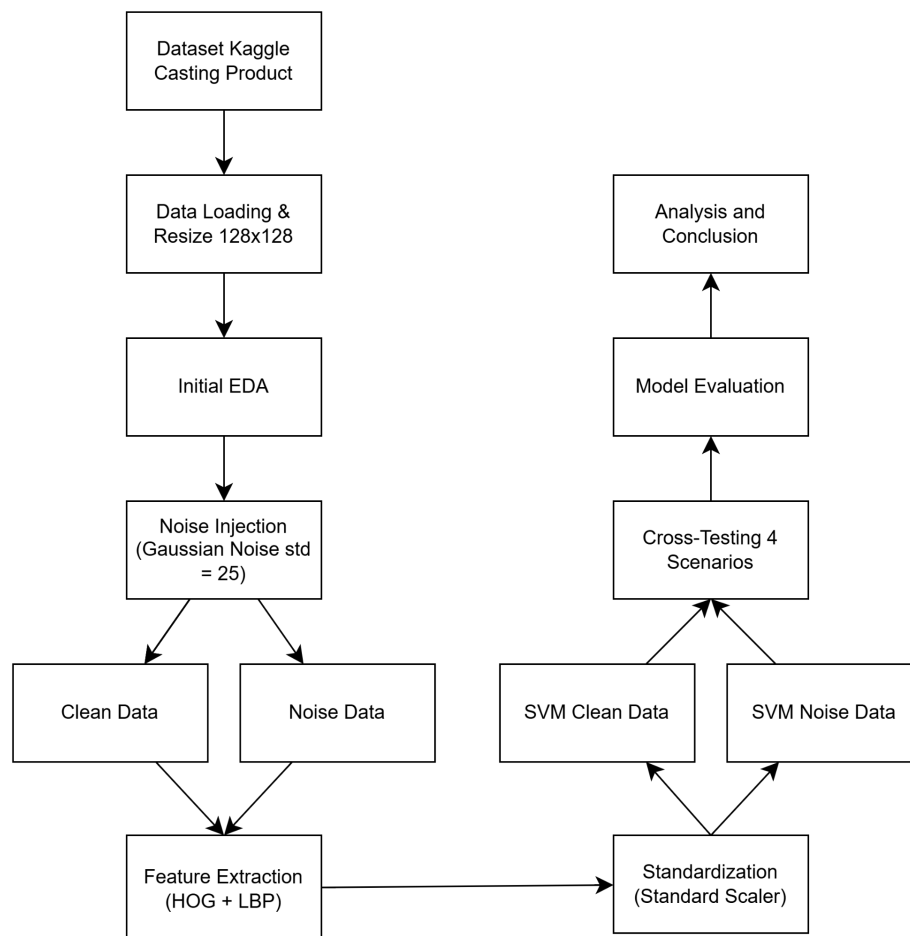


Figure 1. FlowChart Research

Stage 1: Data Loading and Basic Preprocessing

In this stage, all images from the training and test datasets will be loaded from the dataset directory, then each image will be resized to 128x128 pixels. It should be noted that no denoising or augmentation will be performed at this stage, to maintain the original data quality and allow for fair and valid comparisons between scenarios.

Stage 2: Gaussian Noise Injection

To simulate real-world conditions in an industrial environment, image degradation will be performed by adding Gaussian noise to all images using the parameters mean = 0 and standard deviation = 25 [7]. This standard deviation value was chosen because it is considered sufficient to significantly disrupt the image structure without completely eliminating important visual information. This process will produce two parallel versions of the dataset: a clean version and a noisy version, each containing the training and test data.

Stage 3: HOG and LBP Feature Extraction

Before feature extraction, each image will be converted to grayscale because both feature extraction methods require only one color channel. Next, features will be extracted using two methods simultaneously. The first method is the Histogram of Oriented Gradients (HOG), configured with 9 orientations, an 8x8 pixel cell size, and 2x2 cell blocks with L2-Hys normalization. HOG was chosen for its ability to capture edge and shape structure information that represents anomalies on casting surfaces [3]. The second method is the Local Binary Pattern (LBP) with a radius of 3 and 24 sampling points in uniform mode, which will produce a histogram of micro-surface texture patterns and is suitable for detecting subtle variations such as porosity [4]. The feature vectors from both methods will then be combined into a single feature vector per image and standardized using StandardScaler to prevent any single feature from dominating the learning process due to scale differences [5].

Stage 4: SVM Model Training and Overfitting Analysis

In this stage, two Support Vector Machine (SVM) models with RBF kernels will be trained separately. The first model, called the Original Model, will be trained using features from clean training data. The second model, called the Noisy Model, will be trained using features from noisy training data as a form of noise-based augmentation. After training is complete, both models will be evaluated on their respective training data to check for overfitting by comparing accuracy values on the training and test data.

Stage 5: Cross-Testing Evaluation: Four Scenarios

The final stage is a model robustness evaluation, referring to the robustness benchmarking framework introduced by Hendrycks and Dietterich [10], in which model performance is systematically measured against various levels of common degradation, including Gaussian noise. This evaluation will be conducted through four cross-testing scenarios. Scenario 1 tests the Original Model on clean test data as a benchmark for ideal performance. Scenario 2 tests the Original Model on noisy test data to determine the extent of performance degradation due to noise. Scenario 3 tests the Noisy Model on clean test data to ensure that noise augmentation does not impair the model's ability to recognize clean images. Scenario 4 tests the Noisy Model on noisy test data to demonstrate the benefits of augmentation in improving the model's robustness to noise. Each scenario will be evaluated using three metrics, namely accuracy, recall, and F1-score, and is equipped with a confusion matrix to analyze the classification error patterns in more detail.

Stage 6: Analysis and Conclusion

After all evaluation scenarios have been completed, the results from the four scenarios will be summarized and comparatively analyzed in a single summary table containing the accuracy, recall, and F1-score values for each scenario. The analysis will focus on three main questions. First, to what extent does the Original Model experience a performance degradation when faced with noisy data? This will be answered by comparing the results of Scenario 1 and Scenario 2. Second, whether noise-based augmentation successfully improves the model's robustness to noise? This will be answered by comparing the results of Scenario 2 and Scenario 4. Third, whether there is a significant trade-off between robustness to noise and generalization ability under ideal conditions? This will be answered by comparing Scenario 1 and Scenario 3.

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